

Chat Export

Chat Transcript

User: I have an aluminum cooling plate that have 600W of heat to be disapated using six 40mmX20mm fans. What is the min CFM per fan required?

Assistant: To calculate the minimum CFM per fan, I used a practical thermal rule of thumb. For a 600 W load and six fans, the target is roughly 50–100 CFM per fan depending on the allowed temperature rise. A practical recommendation is at least 75–100 CFM per fan.

User: what about the static pressure

Assistant: Static pressure matters because it determines whether the fans can push air through the cooling plate fins. For a cooling plate, typical requirements can range from about 0.5–3 inH₂O depending on the fin density. Your fan's 165 Pa is fairly good, but you still need enough airflow.

User: So the total watt disapated is 432W. I have 6 40mmx20 at 13cfm with 165pa.

Assistant: With 6 fans at 13 CFM each, the total airflow is about 78 CFM. That is roughly the minimum for a 432 W load in an ideal case, but it is fairly marginal once you account for the pressure drop through a real cooling plate.

User: surething. The cooling plate is 11X3.5 inches. total thickness is 0.5inches that includes 0.25 inch fins.

Assistant: With those dimensions, the plate is moderately restrictive. The 0.25 in fin spacing means the fans should be able to push air through it, but the effective airflow will be lower than the free-air rating. The setup is likely borderline rather than comfortably safe.

User: spacing is 0.25inches

Assistant: With 0.25 in spacing, the fin stack is moderately restrictive. The static pressure rating is acceptable, but the airflow is still likely marginal for 432 W. A more comfortable target would be closer to 16–20 CFM per fan, or more fans.

User: yes

Assistant: Using a simple air-heating estimate, 78 CFM total would produce about a 10°C rise in the air temperature in an ideal case. In practice, the plate itself may run noticeably hotter, often around 15–25°C above inlet air depending on flow distribution.

Math included in the discussion:

- Total airflow: $6 \times 13 = 78$ CFM
- Total airflow in SI units: $78 \text{ CFM} \approx 0.0368 \text{ m}^3/\text{s}$
- Air temperature rise estimate: $\Delta T \approx P / (\dot{m} c_p)$
- With air density $\rho \approx 1.2 \text{ kg/m}^3$ and $c_p \approx 1005 \text{ J/(kg}\cdot\text{K)}$:
- $\dot{m} \approx 1.2 \times 0.0368 \approx 0.044 \text{ kg/s}$
- $\Delta T \approx 432 / (0.044 \times 1005) \approx 9.8^\circ\text{C}$
- Practical takeaway: the plate may run about 15–25°C above inlet air.